Chapter 14A

Reproductive Endocrinology and Female Symptom Framework

Original Medvale educational chapter for internal medicine review and Graph-RAG ingestion. Coverage guided by the uploaded Anki export plus outside clinical references; wording is original and explanatory.

Scope of 14A. This segment builds the endocrine and symptom framework for the reproductive chapter. It covers the hypothalamic-pituitary-gonadal axis, ovarian cycle physiology, puberty, menopause, abnormal uterine bleeding, amenorrhea, pelvic pain, vaginal discharge, hirsutism, galactorrhea, and the first-pass approach to female infertility. Disease-specific gynecologic disorders, contraception, pregnancy-aware medicine, infections, malignancy, and male reproductive medicine are handled in Chapters 14B-14D.

Graph-RAG design note. The prose intentionally states causal and diagnostic relationships explicitly, such as “hyperprolactinemia suppresses GnRH,” “chronic anovulation causes unopposed estrogen,” and “pregnancy must be excluded before diagnosing secondary amenorrhea.” These statements are useful for extraction into concept nodes and relationship edges.

High-Yield Relationship Map

Source concept	Relationship	Target concept / clinical consequence
Hypothalamus	pulsatile secretion of	GnRH
GnRH	stimulates	anterior pituitary LH and FSH release
LH in ovary	stimulates	theca cell androgen production
FSH in ovary	stimulates	granulosa aromatase activity and estradiol production
Estradiol late follicular surge	switches from negative to positive feedback and triggers	LH surge and ovulation
Progesterone from corpus luteum	stabilizes	secretory endometrium
Falling progesterone and estradiol	causes	spiral artery constriction and menses
Hyperprolactinemia	suppresses	GnRH pulsatility -> low/normal LH and FSH -> amenorrhea
Chronic anovulation	causes	unopposed estrogen exposure -> endometrial hyperplasia risk
Primary ovarian insufficiency	causes	low estradiol with high FSH before age 40
Functional hypothalamic amenorrhea	causes	low GnRH drive -> low estradiol with low/normal FSH
Pregnancy	must be excluded before	AUB, amenorrhea, pelvic pain, nausea, and breast symptoms are assigned to other causes

1. Reproductive Endocrine Framework

Female reproductive physiology is governed by a feed-forward and feedback loop connecting the hypothalamus, anterior pituitary, ovaries, and endometrium. The hypothalamus releases gonadotropin-releasing hormone (GnRH) in pulses. Pulsatility matters because continuous GnRH exposure suppresses pituitary gonadotropin release, whereas physiologic pulses stimulate luteinizing hormone (LH) and follicle-stimulating hormone (FSH). LH and FSH then coordinate ovarian steroid production, follicle maturation, ovulation, and luteal function. This makes the reproductive axis highly sensitive to stress, low energy availability, hyperprolactinemia, thyroid disease, chronic illness, medications, ovarian failure, and hypothalamic or pituitary disease.

The simplest working model is: hypothalamus produces GnRH; pituitary produces LH and FSH; ovary produces estradiol, progesterone, and inhibin; endometrium responds to ovarian hormones. Disruption at any level can produce irregular menses, amenorrhea, infertility, hot flashes, galactorrhea, androgen excess, or abnormal bleeding. The diagnostic task is to localize the failure point: outflow tract, uterus/endometrium, ovary, pituitary, hypothalamus, or systemic illness.

1.1 The Two-Cell, Two-Gonadotropin Model

The ovary uses two major steroidogenic compartments. Theca cells respond primarily to LH and generate androgens. Granulosa cells respond primarily to FSH and aromatize theca-derived androgens into estradiol. Estradiol supports endometrial proliferation and, at high sustained levels late in the follicular phase, produces positive feedback on the pituitary to generate the LH surge. The LH surge triggers ovulation and luteinization. After ovulation, the corpus luteum produces progesterone, which converts proliferative endometrium into secretory endometrium capable of implantation.

Cell or structure	Dominant stimulus	Main product/action	Clinical connection
Theca cell	LH	Androgen synthesis	Excess theca androgen production contributes to hyperandrogenism in PCOS-like states.
Granulosa cell	FSH	Aromatase activity -> estradiol; inhibin B	Granulosa function is reflected by estradiol production and follicular maturation.
Dominant follicle	FSH support plus estradiol feedback	Selection and maturation	Poor follicular development can cause anovulation and infertility.
Corpus luteum	LH support after ovulation	Progesterone	Luteal progesterone stabilizes secretory endometrium; withdrawal causes menses.
Endometrium	Estradiol then progesterone	Proliferative then secretory transformation	Unopposed estrogen increases hyperplasia risk; progesterone protects endometrium.

1.2 Menstrual Cycle Phases

The menstrual cycle is usually described from the perspective of the ovary and the endometrium. Ovarian phases are follicular, ovulatory, and luteal. Endometrial phases are menstrual, proliferative, and secretory. The follicular phase varies most in length; the luteal phase is more consistent, often roughly two weeks. Therefore, cycle length variation usually reflects differences in time to ovulation rather than major variation in the post-ovulatory luteal interval.

Phase	Dominant biology	Hormonal pattern	Clinical meaning
Early follicular / menstrual	Regression of prior corpus luteum and shedding of endometrium	Low estradiol and progesterone; FSH begins to rise	Cycle day 1 is first day of menses. Low hormones release pituitary from negative feedback.
Mid-late follicular / proliferative	Dominant follicle selection and endometrial growth	Rising estradiol; FSH falls via feedback/inhibin	Estradiol thickens endometrium and produces watery cervical mucus.
Ovulation	Follicle rupture and oocyte release	Sustained high estradiol -> LH surge	LH surge is the immediate endocrine trigger for ovulation.
Luteal / secretory	Corpus luteum function and implantation preparation	Progesterone high; estradiol moderate	Progesterone stabilizes endometrium and raises basal body temperature.
Late luteal	Corpus luteum involution if no pregnancy	Progesterone and estradiol fall	Hormone withdrawal causes menses and may worsen PMS/PMDD symptoms.

Clinical pearl: a patient with regular predictable cycles is very likely ovulating. A patient with oligomenorrhea, skipped cycles, or unpredictable bleeding is more likely to have anovulatory cycles, even if bleeding still occurs periodically.

1.3 Negative Feedback, Positive Feedback, and Axis Localization

Most of the cycle is governed by negative feedback: estradiol, progesterone, and inhibins restrain pituitary gonadotropin output. Late in the follicular phase, sustained high estradiol causes positive feedback and an LH surge. This switch explains why estradiol can both suppress gonadotropins in most settings and trigger LH release in the narrow pre-ovulatory window. Clinically, feedback patterns help localize disease. High FSH with low estradiol implies ovarian failure or reduced ovarian reserve. Low or inappropriately normal FSH with low estradiol implies hypothalamic or pituitary under-stimulation. Normal gonadotropins with hyperandrogenism and chronic oligo-ovulation suggests PCOS after exclusion of mimics.

Pattern	Likely axis location	Examples	Expected reproductive consequence
High FSH, low estradiol	Ovary cannot respond to pituitary signal	Primary ovarian insufficiency, menopause, gonadal dysgenesis, chemotherapy/radiation effect	Amenorrhea, infertility, vasomotor symptoms, low bone density risk
Low/normal FSH, low estradiol	Hypothalamic or pituitary under-stimulation	Functional hypothalamic amenorrhea, pituitary mass, chronic illness, hyperprolactinemia	Amenorrhea, infertility, hypoestrogenic symptoms
High prolactin, low/normal gonadotropins	Pituitary lactotroph effect suppresses GnRH	Prolactinoma, dopamine antagonist medication, hypothyroidism, pregnancy/lactation	Amenorrhea, galactorrhea, infertility
Oligo-ovulation plus androgen excess	Ovarian/adrenal androgen excess after excluding mimics	PCOS, nonclassic CAH, androgen-secreting tumor, Cushing syndrome	Irregular bleeding, infertility, hirsutism, acne

2. Puberty, Menopause, and Life-Stage Anchors

Reproductive complaints should always be interpreted by life stage. A finding that is physiologic in adolescence or perimenopause may be abnormal in a stable reproductive-aged patient. Adolescents commonly have anovulatory cycles in the early years after menarche because the HPG axis is still maturing. Perimenopausal patients can also develop irregular bleeding as ovarian reserve declines and ovulatory cycles become less consistent. However, “expected” life-stage variability does not remove the need to exclude pregnancy, anemia, bleeding disorders, endocrine disease, infection, structural pathology, and malignancy when the presentation is severe, persistent, or high risk.

Life stage	Typical physiology	Common clinical issues	Red flags
Puberty/adolescence	Maturing GnRH pulsatility and early anovulatory cycles	Irregular cycles, dysmenorrhea, acne, heavy menstrual bleeding	No breast development by 13, no menarche by 15 or within 3 years of thelarche, severe anemia, eating disorder signs
Reproductive years	Cyclic ovulation when axis is intact	Pregnancy, contraception, AUB, PCOS, endometriosis, infertility, STIs	Positive pregnancy test with pain/bleeding, hemodynamic instability, fever with pelvic pain, virilization
Perimenopause	Declining ovarian reserve with intermittent anovulation	Irregular bleeding, vasomotor symptoms, sleep/mood changes	Heavy or persistent bleeding, intermenstrual bleeding, risk factors for endometrial cancer
Postmenopause	Permanent loss of ovarian follicular activity; menopause defined after 12 months without menses	Vasomotor symptoms, genitourinary syndrome, bone loss risk	Any postmenopausal bleeding is abnormal until evaluated.

Menopause is a retrospective clinical diagnosis after 12 months without menses in the absence of another pathologic or physiologic cause. The key endocrine pattern is ovarian follicular depletion, reduced estradiol, and loss of ovarian negative feedback, producing elevated FSH. Vasomotor symptoms, sleep disturbance, mood symptoms, vaginal dryness, dyspareunia, and recurrent urinary symptoms can all be connected to estrogen withdrawal. Postmenopausal bleeding is never considered normal and requires evaluation for endometrial cancer or other pathology.

3. Female Symptom Framework: First Questions Before Disease Labels

Female reproductive complaints are easiest to evaluate when approached as syndromes rather than as isolated diagnoses. The first branch points are pregnancy status, hemodynamic stability, infection signs, malignancy risk, and whether the problem is bleeding, pain, discharge, absent menses, androgen excess, breast/galactorrhea symptoms, or infertility. Pregnancy testing is central because pregnancy changes the differential diagnosis, the urgency of pelvic pain or bleeding, and the safety of medications and imaging.

Symptom	First “must not miss” categories	Initial tests or actions	Key relationship
Amenorrhea	Pregnancy, pituitary disease, ovarian insufficiency, hypothalamic suppression	Pregnancy test; TSH, prolactin, FSH/LH/estradiol; targeted androgen testing	Amenorrhea localizes to pregnancy, outflow tract, uterus, ovary, pituitary, hypothalamus, or systemic illness.

Symptom	First “must not miss” categories	Initial tests or actions	Key relationship
Abnormal uterine bleeding	Pregnancy complication, anemia, coagulopathy, endometrial cancer	Pregnancy test, CBC; consider TSH, coagulopathy tests, STI tests, ultrasound, biopsy by risk	Anovulation causes unopposed estrogen; unopposed estrogen causes endometrial proliferation and irregular bleeding.
Pelvic pain	Ectopic pregnancy, torsion, PID/TOA, appendicitis, ruptured cyst	Pregnancy test, vitals, pelvic exam, CBC/UA/STI testing, ultrasound; urgent evaluation if unstable	Pain plus positive pregnancy test is ectopic pregnancy until proven otherwise.
Vaginal discharge	Cervicitis, PID, STI, vaginitis	pH, wet mount/NAATs when available, pregnancy test if relevant, STI testing by risk	Vaginitis usually causes local vaginal symptoms; cervicitis/PID involve ascending infection risk.
Hirsutism/virilization	PCOS, CAH, Cushing syndrome, androgen-secreting tumor	Total/free testosterone, DHEA-S, 17-hydroxyprogesterone when indicated	Rapid virilization suggests tumor until excluded.
Infertility	Anovulation, tubal disease, uterine disease, male factor	Evaluate both partners; ovulation assessment, semen analysis, tubal/uterine assessment	Infertility is a couple-level diagnosis, not only a female diagnosis.

4. Abnormal Uterine Bleeding

Abnormal uterine bleeding (AUB) refers to bleeding that is abnormal in volume, regularity, duration, or timing. It includes heavy menstrual bleeding, intermenstrual bleeding, prolonged bleeding, frequent bleeding, infrequent bleeding, and postmenopausal bleeding. In reproductive-aged patients, AUB must be separated from pregnancy-related bleeding before assigning a gynecologic label. A pregnancy test is therefore a first-line test in any patient with reproductive potential and abnormal bleeding.

The PALM-COEIN framework separates structural causes from nonstructural causes. PALM represents polyp, adenomyosis, leiomyoma, and malignancy/hyperplasia. COEIN represents coagulopathy, ovulatory dysfunction, endometrial, iatrogenic, and not otherwise classified. This classification is useful for Graph-RAG because it creates edges between bleeding patterns and mechanisms. For example, leiomyomas distort uterine anatomy, ovulatory dysfunction causes unstable endometrial proliferation, anticoagulants cause iatrogenic bleeding, and chronic anovulation increases hyperplasia risk through unopposed estrogen.

PALM-COEIN category	Mechanism	Typical clues	Evaluation angle
Polyp	Focal endometrial or cervical overgrowth	Intermenstrual bleeding, spotting	Transvaginal ultrasound, saline infusion sonography, hysteroscopy when needed
Adenomyosis	Endometrial tissue within myometrium	Heavy painful menses, enlarged tender uterus	Ultrasound or MRI when uncertain
Leiomyoma	Benign smooth muscle tumor; submucosal lesions bleed most	Heavy bleeding, bulk symptoms, infertility depending on location	Ultrasound first-line
Malignancy/hyperplasia	Unopposed estrogen, genetic risk, age-related risk	Postmenopausal bleeding, persistent AUB, obesity/PCOS risk	Endometrial sampling by age/risk
Coagulopathy	Impaired hemostasis	Heavy bleeding since menarche, bruising, epistaxis, family history	CBC, ferritin, coagulation/vWD testing when suggested
Ovulatory dysfunction	Irregular estrogen exposure and absent luteal progesterone	Oligomenorrhea, unpredictable bleeding, PCOS/thyroid/prolactin clues	Pregnancy test, TSH, prolactin, androgen evaluation, ovulation history
Endometrial	Primary local endometrial hemostasis disorder	Regular cycles with heavy bleeding after other causes excluded	Diagnosis of exclusion
Iatrogenic	Medication or device effect	Anticoagulants, hormonal therapy, IUD-related bleeding	Medication/device review

AUB evaluation should ask whether the patient is unstable, pregnant, anemic, infected, anticoagulated, or at risk for endometrial cancer. Acute heavy bleeding with hypotension, syncope, severe anemia, or soaking through pads rapidly is a stabilization problem before it is a diagnostic puzzle. Chronic AUB requires attention to iron deficiency, quality of life, contraception goals, pregnancy goals, cancer risk, and contraindications to hormonal therapy.

High-yield endometrial sampling rule: endometrial tissue sampling is generally recommended for AUB in patients age 45 years or older and for younger patients with significant risk factors for unopposed estrogen exposure, persistent bleeding,

or failed medical management. Risk factors include chronic anovulation, obesity, PCOS, Lynch syndrome, tamoxifen exposure, and prolonged irregular bleeding.

AUB first-pass algorithm	Diagnostic intent
1. Check vital signs and bleeding severity.	Identify unstable bleeding that requires immediate stabilization.
2. Perform pregnancy test in reproductive potential.	Exclude intrauterine pregnancy complications and ectopic pregnancy.
3. Obtain CBC; consider ferritin if chronic/heavy.	Measure anemia and iron deficiency burden.
4. Decide whether infection testing, TSH, prolactin, androgen testing, or coagulopathy testing is indicated.	Use history and exam to connect bleeding to endocrine, infectious, or hemostatic mechanisms.
5. Use pelvic ultrasound when structural pathology is suspected or bleeding persists.	Evaluate fibroids, polyps, adenomyosis, ovarian pathology, endometrial thickness/pattern.
6. Perform endometrial sampling by age/risk.	Exclude hyperplasia or malignancy when risk is nontrivial.

5. Amenorrhea and Oligomenorrhea

Amenorrhea is the absence of menses when menses should be present. Primary amenorrhea is generally evaluated when there is no menarche by age 15 or within 3 years after thelarche, or earlier when pubertal development is absent or abnormal. Secondary amenorrhea is absence of menses for at least 3 months in someone with previously regular cycles or at least 6 months in someone with previously irregular cycles. Oligomenorrhea refers to infrequent menses and often reflects oligo-ovulation. The clinical meaning of amenorrhea depends on whether the issue is pregnancy, outflow obstruction, absent uterus, ovarian insufficiency, hypothalamic/pituitary suppression, endocrine disease, or androgen excess.

Pregnancy is the most important initial diagnosis to exclude. After pregnancy is excluded, the initial laboratory backbone often includes TSH, prolactin, FSH, LH, and estradiol, individualized by the clinical scenario. TSH is included because thyroid disease can disrupt ovulatory function and prolactin dynamics. Prolactin is included because prolactin suppresses GnRH pulsatility and can cause amenorrhea, galactorrhea, and infertility. FSH and estradiol help distinguish ovarian failure from central under-stimulation. Androgen testing is added when hirsutism, acne, virilization, or oligomenorrhea suggests hyperandrogenism.

Amenorrhea pattern	Core mechanism	Typical labs	Common causes
Pregnancy	hCG maintains corpus luteum and suppresses new cycle	Positive hCG	Intrauterine pregnancy, ectopic pregnancy, pregnancy loss
Outflow tract or uterine cause	Endometrium cannot shed or blood cannot exit	Hormones may be normal	Mullerian agenesis, imperforate hymen, transverse vaginal septum, Asherman syndrome
Primary ovarian insufficiency	Ovary fails despite pituitary stimulation	High FSH, low estradiol	Autoimmune, genetic/FMR1 premutation, Turner mosaicism, chemotherapy/radiation, idiopathic
Functional hypothalamic amenorrhea	Low energy/stress suppresses GnRH	Low/normal FSH/LH, low estradiol	Weight loss, eating disorder, intense exercise, stress, chronic illness
Hyperprolactinemia	Prolactin suppresses GnRH	High prolactin; low/normal gonadotropins	Prolactinoma, pregnancy/lactation, dopamine blockers, hypothyroidism, renal failure
PCOS / chronic anovulation	Oligo-ovulation plus androgen excess	Androgens may be high; LH/FSH ratio is not required	PCOS after exclusion of mimics

Amenorrhea localization can be simplified by asking: Is the patient pregnant? Is there uterus/outflow tract anatomy capable of bleeding? Is estrogen present? Is FSH high or low? Is prolactin high? Is androgen excess present? These questions convert a broad list of diagnoses into a small set of anatomic and endocrine locations.

5.1 Primary Amenorrhea

Primary amenorrhea has two major branches: normal secondary sexual characteristics versus absent or incomplete secondary sexual characteristics. Normal breast development indicates prior estrogen exposure. If breast development is normal but menses never occurred, consider outflow tract obstruction or absent uterus, including imperforate hymen, transverse vaginal septum, Mullerian agenesis, or androgen insensitivity depending on anatomy and karyotype. Absent breast development suggests hypoestrogenism from ovarian failure, gonadal dysgenesis, hypothalamic disease, pituitary disease, or constitutional delay.

Primary amenorrhea clue	Mechanistic meaning	Examples
Breasts present, uterus absent	Estrogen exposure occurred; uterus did not develop or is not present	Mullerian agenesis; complete androgen insensitivity has absent uterus with scant/absent pubic hair and 46,XY karyotype
Breasts present, uterus present, cyclic pain	Estrogenized endometrium with blocked outflow	Imperforate hymen, transverse vaginal septum
Breasts absent, high FSH	Primary gonadal failure	Turner syndrome/gonadal dysgenesis, primary ovarian insufficiency
Breasts absent, low/normal FSH	Central under-stimulation	Functional hypothalamic amenorrhea, pituitary disease, constitutional delay

5.2 Secondary Amenorrhea

Secondary amenorrhea is most commonly approached with pregnancy testing first, then targeted endocrine evaluation. A classic high-yield sequence is pregnancy test, TSH, prolactin, FSH, LH, and estradiol, followed by androgen testing, pelvic ultrasound, pituitary imaging, or uterine/outflow evaluation when suggested. The most common buckets are pregnancy, PCOS/chronic anovulation, hypothalamic amenorrhea, hyperprolactinemia, thyroid disease, primary ovarian insufficiency, medications, and uterine scarring.

Step	Question	Interpretation
1	Is hCG positive?	Pregnancy-related amenorrhea. If pain or bleeding is present, ectopic pregnancy must be considered.
2	Is prolactin elevated?	Evaluate medications, pregnancy/lactation, hypothyroidism, renal disease, and pituitary adenoma depending on degree/context.
3	Is TSH abnormal?	Hypothyroidism or hyperthyroidism can disrupt ovulation and menstrual regularity.
4	Is FSH high with low estradiol?	Primary ovarian insufficiency or menopause physiology. In patients under 40, evaluate for POI and long-term estrogen deficiency consequences.
5	Is FSH low/normal with low estradiol?	Central under-stimulation: hypothalamic, pituitary, stress/energy deficit, chronic illness, medications.
6	Are androgen signs present?	Evaluate PCOS and mimics; rapid virilization or very high androgens suggest tumor.
7	Did amenorrhea follow uterine instrumentation or infection?	Consider Asherman syndrome or structural endometrial/outflow pathology.

6. Hyperandrogenism, Hirsutism, and Virilization

Hyperandrogenism is a clinical and biochemical bridge between reproductive endocrinology, dermatology, adrenal disease, ovarian disease, and metabolic medicine. Hirsutism means terminal hair growth in androgen-sensitive areas. Acne and androgenic alopecia can also reflect androgen effect. Virilization is more concerning and includes deepening voice, clitoromegaly, rapidly progressive muscle mass changes, male-pattern balding, or severe rapidly progressive hirsutism. Virilization suggests a high androgen burden and should trigger evaluation for androgen-secreting ovarian or adrenal tumors.

PCOS is common, but it should not become a reflex diagnosis before excluding pregnancy, thyroid disease, hyperprolactinemia, nonclassic congenital adrenal hyperplasia, Cushing syndrome, androgen-secreting tumors, and medication exposure when indicated. The 2023 international PCOS guideline supports diagnosis in adults using updated Rotterdam-based criteria: oligo/anovulation, clinical or biochemical hyperandrogenism, and polycystic ovarian morphology or elevated AMH, after exclusion of other causes. In adolescents, diagnostic caution is required because irregular cycles and acne can be physiologic during pubertal maturation.

Finding	Suggests	Next diagnostic thought
Slow hirsutism plus oligomenorrhea	PCOS phenotype likely	Assess metabolic risk; exclude mimics based on presentation.
Rapid virilization	Androgen-secreting tumor until excluded	Check testosterone/DHEA-S urgently and image ovary/adrenal as indicated.
High DHEA-S predominance	Adrenal androgen source	Consider adrenal tumor or adrenal hyperplasia.

Finding	Suggests	Next diagnostic thought
Elevated 17-hydroxyprogesterone	Nonclassic CAH screen positive	Confirm with appropriate endocrine testing.
Cushingoid features	Cortisol excess	Test for Cushing syndrome rather than labeling PCOS.
Galactorrhea or headaches/visual symptoms	Prolactin/pituitary disorder	Check prolactin; pituitary imaging when indicated.

7. Galactorrhea and Hyperprolactinemia

Galactorrhea is milk-like nipple discharge unrelated to normal lactation. Hyperprolactinemia can cause galactorrhea, amenorrhea, oligomenorrhea, infertility, low libido, and hypoestrogenic symptoms because prolactin suppresses GnRH pulsatility. This reduces pituitary LH/FSH stimulation and disrupts ovarian steroid production and ovulation. Not all galactorrhea is pathologic, and not all hyperprolactinemia is a prolactinoma; pregnancy, lactation, nipple stimulation, hypothyroidism, renal disease, chest wall injury, and medications are common causes.

Cause category	Examples	Mechanism
Physiologic	Pregnancy, lactation, nipple stimulation, stress	Normal or transient prolactin elevation.
Medication	Dopamine antagonists, antipsychotics, metoclopramide, some antidepressants, opioids	Reduced dopamine inhibition of lactotrophs.
Endocrine	Primary hypothyroidism	TRH can stimulate prolactin secretion; thyroid replacement may correct prolactin.
Pituitary	Prolactinoma, stalk effect from mass	Autonomous prolactin secretion or reduced dopamine delivery.
Systemic	Chronic kidney disease, cirrhosis	Reduced clearance and altered endocrine regulation.

Clinical pearl: prolactin elevation plus amenorrhea is not only a pituitary diagnosis. Always review pregnancy status, medications, thyroid function, renal function, and symptoms of mass effect. Headache, visual field deficits, cranial nerve symptoms, or very high prolactin levels increase concern for pituitary pathology.

8. Pelvic Pain Framework

Pelvic pain should be triaged before being categorized. The first questions are whether the patient is pregnant, unstable, febrile, peritoneal, or at risk for ovarian torsion or ectopic pregnancy. Acute pelvic pain with a positive pregnancy test is ectopic pregnancy until proven otherwise. Acute unilateral pelvic pain with nausea/vomiting and an adnexal mass suggests ovarian torsion. Pelvic pain with fever, cervical motion tenderness, uterine/adnexal tenderness, and STI risk suggests PID. Severe pain with hemodynamic instability can reflect ruptured ectopic pregnancy, hemorrhagic cyst rupture, or intra-abdominal bleeding.

Pain pattern	Dangerous diagnosis	Helpful clues	Initial direction
Pain + positive pregnancy test	Ectopic pregnancy	Bleeding, unilateral pain, shoulder pain, syncope, adnexal tenderness	Urgent ultrasound and quantitative hCG strategy; stabilize if unstable.
Sudden unilateral pain + nausea/vomiting	Ovarian torsion	Adnexal mass, intermittent severe pain	Urgent gynecology evaluation; Doppler flow does not fully exclude torsion.
Pelvic pain + fever/discharge/CMT	PID or tubo-ovarian abscess	Cervicitis, STI risk, adnexal/uterine tenderness	Pregnancy test, STI testing, empiric treatment when clinical criteria met.
Cyclic chronic pain	Endometriosis or adenomyosis	Dysmenorrhea, dyspareunia, infertility; enlarged tender uterus in adenomyosis	Symptom pattern, exam, ultrasound for structural disease; laparoscopy for definitive endometriosis when needed.
Midcycle brief pain	Mittelschmerz	Ovulatory timing, self-limited unilateral pain	Diagnosis of exclusion when benign and recurrent.
Pelvic pain with urinary/GI symptoms	Non-gynecologic mimics	Dysuria, hematuria, bowel changes, appendiceal signs	UA/culture, pregnancy test, targeted imaging/labs.

9. Vaginal Discharge and Lower Genital Tract Symptoms

Vaginal discharge should be divided into vaginitis, cervicitis, and upper genital tract infection. Vaginitis usually produces vaginal irritation, odor, pruritus, burning, dyspareunia, or discharge. Cervicitis can produce mucopurulent cervical discharge, postcoital bleeding, intermenstrual bleeding, or dysuria, and may be asymptomatic. PID represents ascending infection involving the uterus, fallopian tubes, ovaries, or pelvic peritoneum; it matters because PID can cause infertility, ectopic pregnancy, chronic pelvic pain, and tubo-ovarian abscess.

Syndrome	Typical features	Mechanistic relationship	Testing direction
Bacterial vaginosis	Thin gray/white discharge, fishy odor, elevated pH	Loss of lactobacilli with polymicrobial anaerobic overgrowth	Vaginal pH, whiff test, clue cells, NAAT where available
Vulvovaginal candidiasis	Pruritus, erythema, thick curdy discharge, normal pH	Candida overgrowth/inflammation; not classically an STI	Wet mount/KOH, culture/NAAT if recurrent or complicated
Trichomoniasis	Frothy discharge, odor, irritation, elevated pH, strawberry cervix possible	Sexually transmitted protozoal infection	NAAT preferred when available; treat partners
Cervicitis	Mucopurulent discharge, friable cervix, postcoital bleeding	Often chlamydia or gonorrhea; can ascend to PID	NAAT for chlamydia/gonorrhea; evaluate STI risks
PID	Pelvic/lower abdominal pain, CMT, uterine/adnexal tenderness, fever/discharge possible	Ascending infection injures tubes and pelvic structures	Clinical diagnosis; pregnancy test; STI testing; treat promptly

A practical distinction is that vaginitis is primarily a vaginal ecology/inflammation problem, cervicitis is a cervical STI/inflammation problem, and PID is an ascending upper-genital-tract infection. The same pathogen can appear in more than one syndrome; chlamydia and gonorrhea can cause cervicitis and can ascend to PID.

10. Female Infertility: First-Pass Internal Medicine Approach

Infertility is usually defined as inability to achieve pregnancy after 12 months of regular unprotected intercourse, or after 6 months when the female partner is age 35 or older. Earlier evaluation is appropriate when there is known anovulation, amenorrhea, severe endometriosis, prior pelvic infection, chemotherapy exposure, suspected tubal disease, recurrent pregnancy loss, or male factor concerns. The key clinical principle is that infertility is a couple-level condition: evaluation should include ovulation, ovarian reserve when relevant, tubal/uterine anatomy, and semen analysis rather than assuming a female-only cause.

Domain	Question	Common tools	Graph-RAG relationship
Ovulation	Is an oocyte being released predictably?	Cycle history, mid-luteal progesterone, ovulation predictor kits	Regular cycles suggest ovulation; oligomenorrhea suggests oligo-ovulation.
Ovarian reserve	Is follicular pool reduced?	Age, AMH, day-3 FSH/estradiol, antral follicle count	Reduced reserve lowers response to stimulation but does not alone prove infertility.
Tubal factor	Can sperm and oocyte meet?	History of PID/ectopic/endometriosis; hysterosalpingography	PID damages fallopian tubes and increases infertility/ectopic risk.
Uterine factor	Can implantation and pregnancy be supported?	Ultrasound, saline infusion sonography, hysteroscopy	Submucosal fibroids, polyps, adhesions can impair implantation or cause loss.
Male factor	Is semen adequate for fertilization?	Semen analysis	Male factor contributes to many infertile couples and should be checked early.
Systemic/endocrine	Is there thyroid, prolactin, metabolic, or medication effect?	TSH, prolactin, A1c/metabolic review, medication review	Systemic endocrine disease can impair ovulation and pregnancy health.

11. Integrated Diagnostic Algorithms

Algorithm 1: Secondary Amenorrhea

Branch	Interpretation and next step
Pregnancy test positive	Treat as pregnancy-related until location and viability are clarified. Pain or bleeding requires ectopic pregnancy evaluation.
Pregnancy test negative + high prolactin	Review medications and pregnancy/lactation history; check TSH and renal function; consider pituitary MRI when persistent/unexplained or with mass-effect symptoms.

Branch	Interpretation and next step
Pregnancy test negative + abnormal TSH	Treat thyroid disease and reassess reproductive function.
High FSH + low estradiol	Ovarian insufficiency/menopause pattern. If under age 40, evaluate for primary ovarian insufficiency and counsel on bone/cardiovascular/fertility implications.
Low/normal FSH + low estradiol	Central hypogonadism. Evaluate energy deficit, exercise, stress, chronic illness, pituitary disease, medications, eating disorder risk.
Androgen excess + oligo-ovulation	Consider PCOS after excluding mimics; evaluate metabolic risk and endometrial protection needs.
History of uterine instrumentation + low/absent bleeding despite hormones	Consider intrauterine adhesions/Asherman syndrome.

Algorithm 2: Abnormal Uterine Bleeding

Branch	Interpretation and next step
Unstable or severe acute bleeding	Stabilize first: vitals, IV access, CBC/type and screen when appropriate, urgent gynecology involvement if needed.
Pregnancy test positive	Pregnancy complication pathway: ectopic pregnancy, miscarriage, implantation bleeding, gestational trophoblastic disease depending on context.
Age >=45 or high-risk younger patient	Endometrial sampling to evaluate hyperplasia/malignancy.
Regular heavy menses since menarche	Consider coagulopathy, especially von Willebrand disease, platelet disorder, or anticoagulant effect.
Irregular unpredictable bleeding with oligomenorrhea	Ovulatory dysfunction; evaluate PCOS, thyroid disease, prolactin, hypothalamic causes, medication effects.
Bulk symptoms or enlarged uterus	Structural pathway: fibroid, adenomyosis, pregnancy, malignancy; ultrasound is usually first-line.

Algorithm 3: Acute Pelvic Pain

Branch	Interpretation and next step
Positive pregnancy test	Ectopic pregnancy must be excluded; stabilize immediately if hypotension, syncope, peritoneal signs, or severe bleeding.
Sudden unilateral pain with nausea/vomiting	Ovarian torsion pathway; urgent gynecology consultation.
Fever, cervical motion tenderness, uterine/adnexal tenderness, discharge	PID pathway; test for pregnancy and STIs, treat promptly when clinically suspected.
Peritoneal signs, RLQ pain, GI symptoms	Do not anchor on gynecologic diagnoses; evaluate appendicitis, diverticulitis, bowel disease, urinary stones, pyelonephritis.
Recurrent cyclic pain	Consider endometriosis, adenomyosis, dysmenorrhea, pelvic floor dysfunction, bladder pain syndrome, IBS overlap.

12. High-Yield Comparison Tables

Amenorrhea Lab Patterns

Diagnosis bucket	FSH	Estradiol	Prolactin	Key clue
Pregnancy	Variable/suppressed	High relative to nonpregnant state	May rise	Positive hCG
Functional hypothalamic amenorrhea	Low/normal	Low	Normal or mildly altered	Low weight, energy deficit, exercise, stress
Primary ovarian insufficiency	High	Low	Normal	Age under 40 with ovarian failure pattern
Hyperprolactinemia	Low/normal	Low	High	Galactorrhea, medication, pituitary symptoms
PCOS/chronic anovulation	Often normal	Often normal or mildly altered	Normal	Oligomenorrhea plus androgen excess
Asherman/outflow issue	Often normal	Often normal	Normal	Amenorrhea after uterine instrumentation or obstructive symptoms

Dangerous Diagnoses Not to Miss

Presentation	Must-not-miss diagnosis	Why it matters
Pelvic pain + positive hCG	Ectopic pregnancy	Can rupture and cause life-threatening hemorrhage.
Sudden unilateral pelvic pain + vomiting	Ovarian torsion	Ovary can become ischemic; time-sensitive surgical diagnosis.
Postmenopausal bleeding	Endometrial cancer until evaluated	Bleeding after menopause is abnormal and requires evaluation.
AUB age >=45 or high-risk younger patient	Endometrial hyperplasia/cancer	Chronic unopposed estrogen increases malignant potential.
Amenorrhea + headache/visual field symptoms	Pituitary mass	Mass effect can threaten vision and indicate macroadenoma or other lesion.
Rapid virilization	Androgen-secreting tumor	Rapid/severe androgen excess is not typical uncomplicated PCOS.
Pelvic pain + fever + adnexal mass	Tubo-ovarian abscess	Can rupture, cause sepsis, and impair fertility.

13. Chapter 14A Summary for Graph-RAG Extraction

The reproductive axis is a pulse-dependent endocrine network. GnRH pulse frequency and amplitude drive pituitary LH and FSH. LH and FSH drive ovarian androgen, estradiol, progesterone, inhibin, follicular maturation, ovulation, and luteal function. Estradiol proliferates endometrium; progesterone stabilizes and differentiates endometrium; withdrawal of progesterone and estradiol causes menses. Chronic anovulation removes regular progesterone exposure, allowing unopposed estrogen to stimulate endometrial proliferation and raising hyperplasia risk.

Amenorrhea is not a single disease. It is a symptom caused by pregnancy, outflow tract obstruction, uterine/endometrial injury, ovarian failure, pituitary disease, hypothalamic suppression, systemic illness, thyroid disease, prolactin excess, androgen excess, or medication effects. A pregnancy test is the first diagnostic step in reproductive-aged amenorrhea. Prolactin, TSH, FSH, LH, and estradiol help localize the level of dysfunction. High FSH with low estradiol points to ovarian insufficiency. Low or inappropriately normal FSH with low estradiol points to hypothalamic or pituitary under-stimulation. High prolactin suppresses GnRH and can cause amenorrhea and galactorrhea.

Abnormal uterine bleeding should be classified using PALM-COEIN. Structural causes include polyp, adenomyosis, leiomyoma, and malignancy/hyperplasia. Nonstructural causes include coagulopathy, ovulatory dysfunction, endometrial causes, iatrogenic causes, and not otherwise classified causes. Endometrial sampling is particularly important in patients age 45 or older and in younger patients with risk factors for unopposed estrogen or persistent bleeding. Pelvic pain requires immediate exclusion of ectopic pregnancy, ovarian torsion, PID/tubo-ovarian abscess, and non-gynecologic emergencies. Vaginal discharge should be sorted into vaginitis, cervicitis, and PID because those syndromes differ in complications and treatment implications.

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